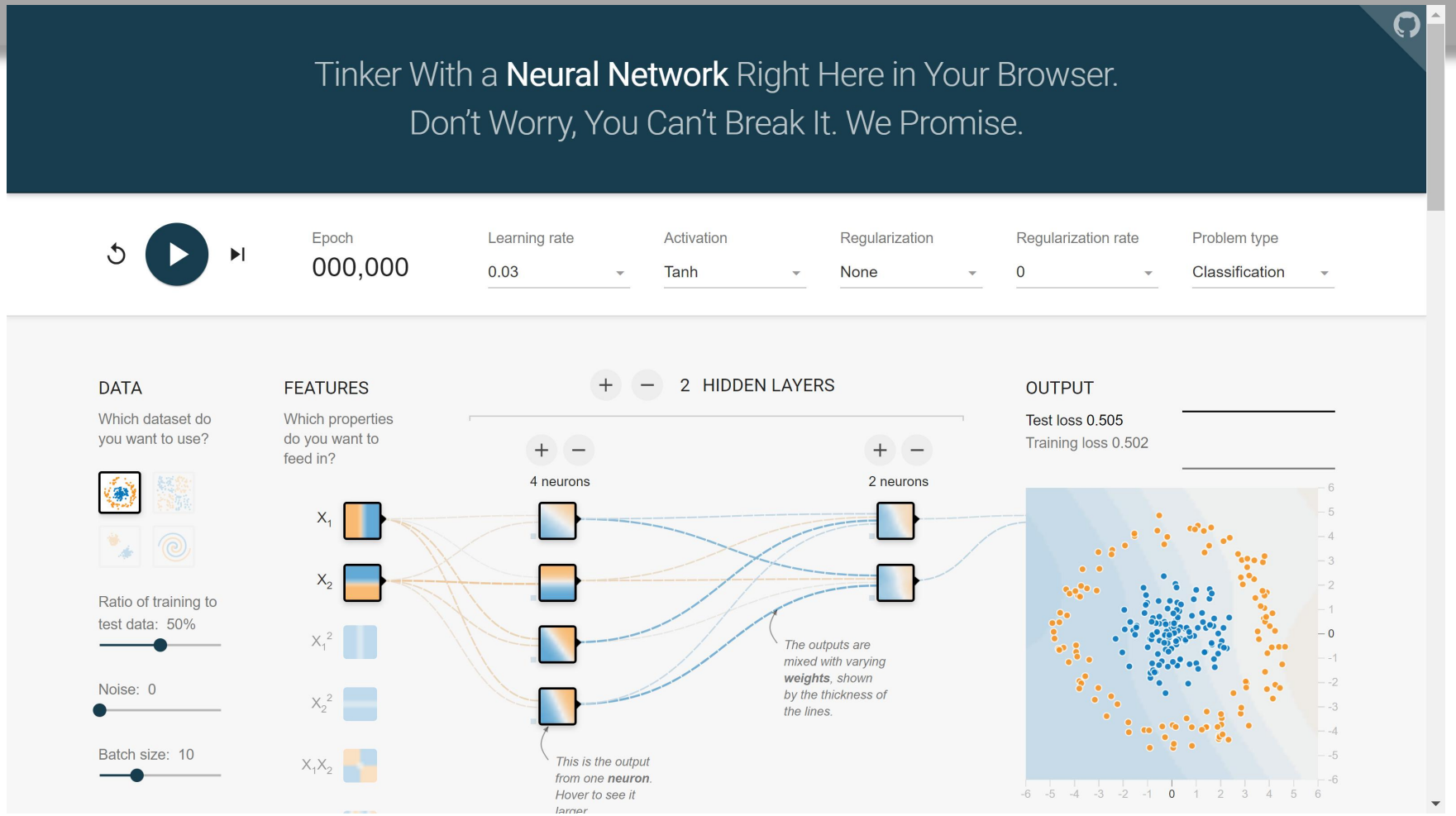
CS5187: Computer Vision

Neural networks



New problem: Image classification

- Input: an image
  - Assume it is of fixed size, e.g. 128x128 pixels
- Output: the class label for that image
- Label is generally a discrete label from a pre-defined set
  - e.g. {cat, dog, cow, toaster, apple, tomato, truck, … }

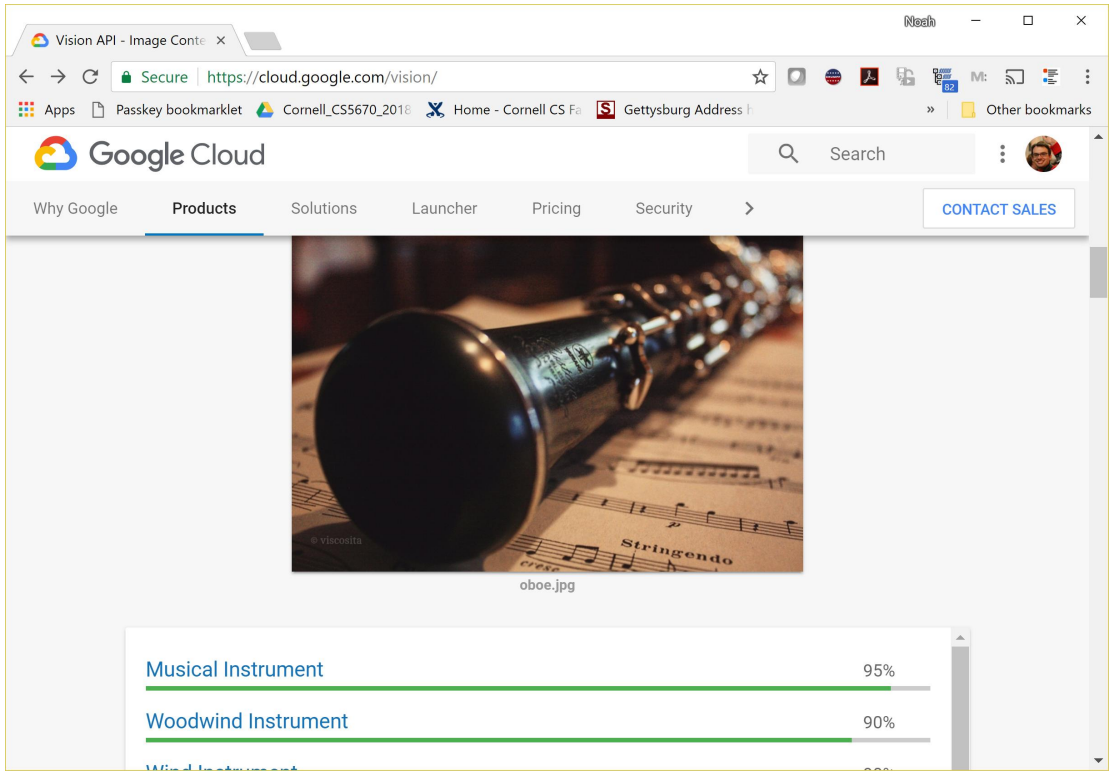
```
def classifier(image):
    //Do some stuff
    return class_label;
```

$f($  $) =$ "Cat"

$f($  $) =$ "Dog"

$f($  $) =$ "Toaster"

Image classification demo



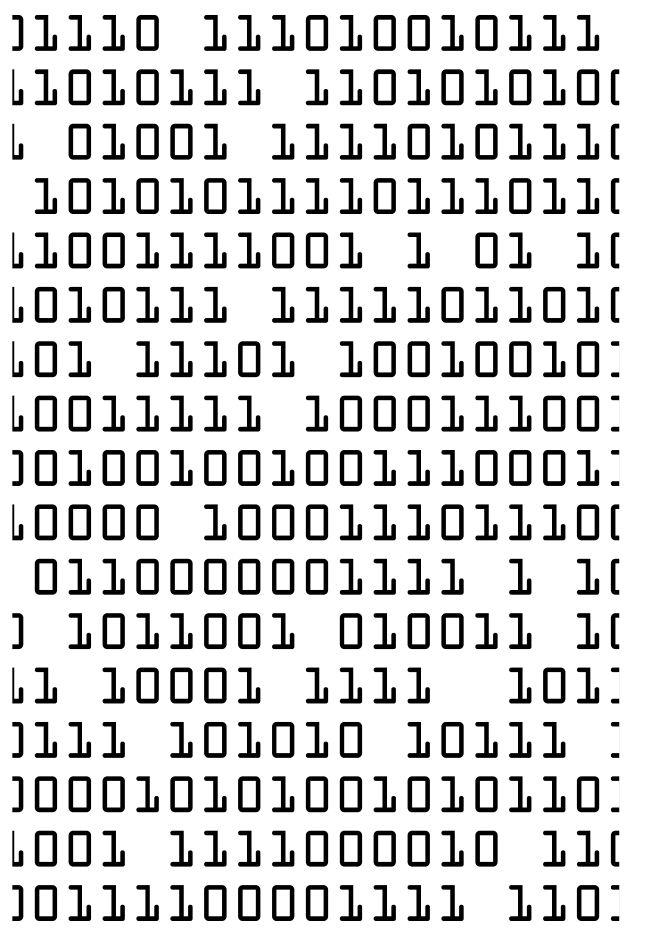
https://cloud.google.com/vision/docs/drag-and-drop

See also:
https://aws.amazon.com/rekognition/
https://www.clarifai.com/
https://azure.microsoft.com/en-us/services/cognitive-services/computer-vision/
…

The Semantic Gap



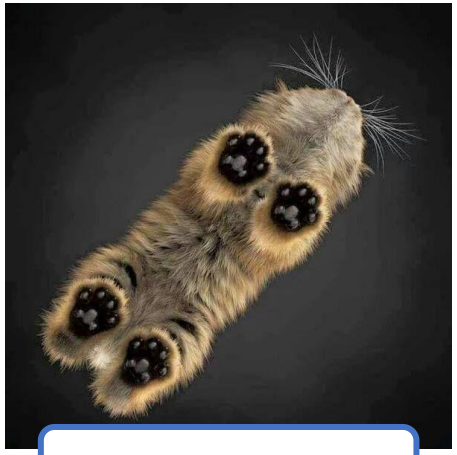
What we see



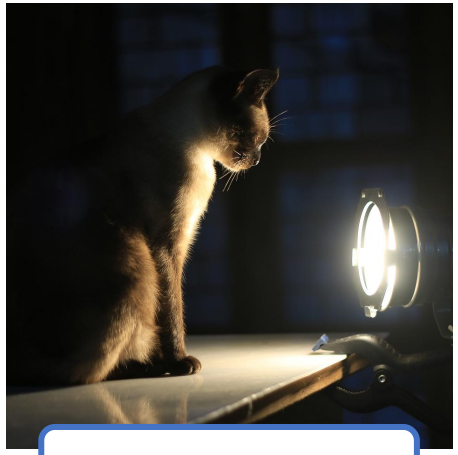
What the computer sees

Variation Makes Recognition Hard

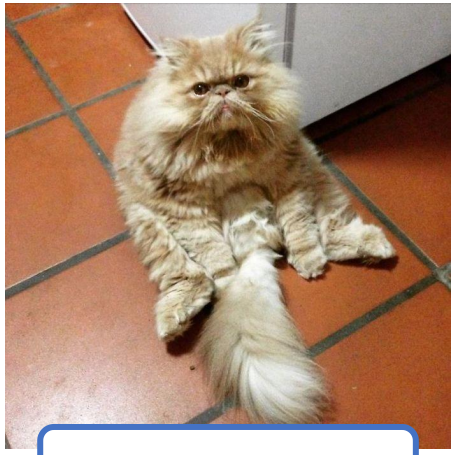
- The same class of object can appear *very* differently in different images



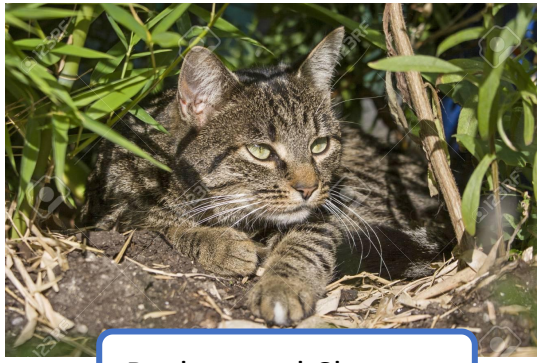
Viewpoint Variation



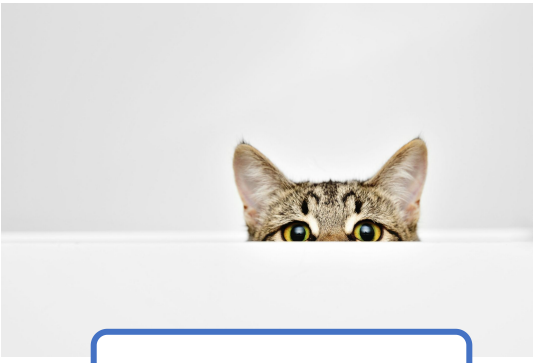
Lighting Variation



Deformation



Background Clutter



Occlusion

The Problem is Under-constrained

- Distinct realities can produce the same image…
- We generally can't compute the "right" answer, but we can compute the most likely one…
- We need some kind of prior to condition on. We can learn this prior from data:

$$f(x) = \underset{\ell_x}{argmax}\ P(\ell_x | data)$$





Images As High-Dimensional Vectors

- An image is just a bunch of numbers
- Let's stack them up into a vector
  - Our training data is just a bunch of high-dimensional points now



Images As High-Dimensional Vectors

- An image is just a bunch of numbers
- Let's stack them up into a vector
  - Our training data is just a bunch of high-dimensional points now

- Divide space into different regions for different classes

